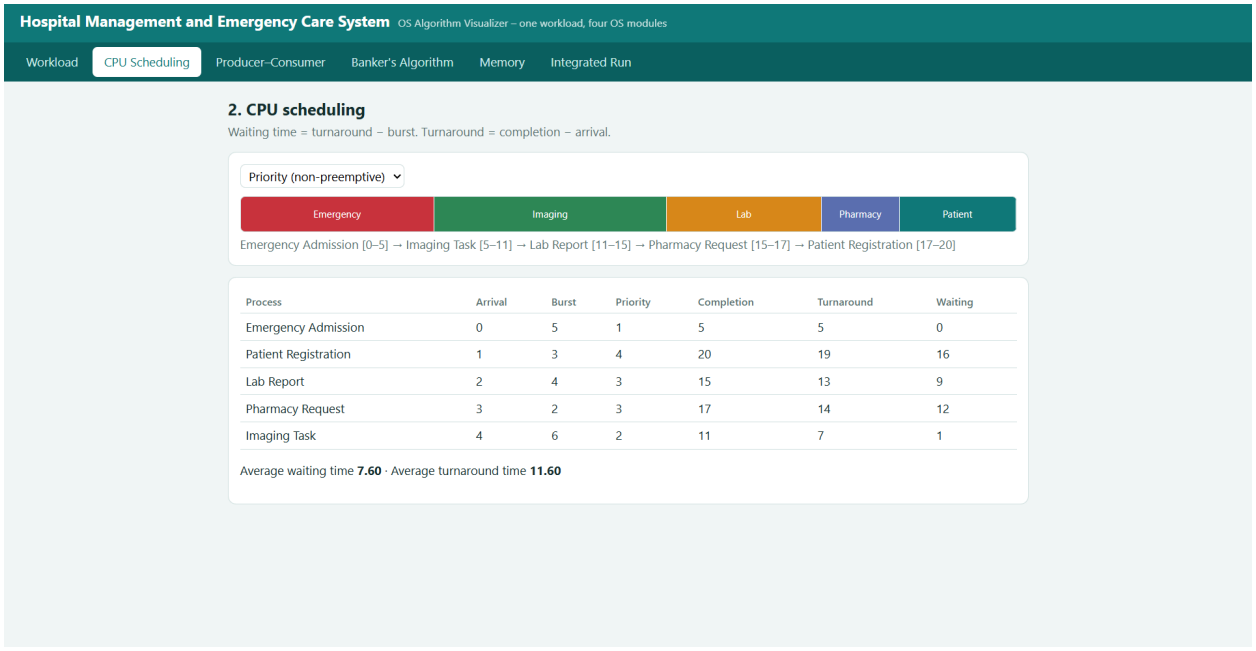
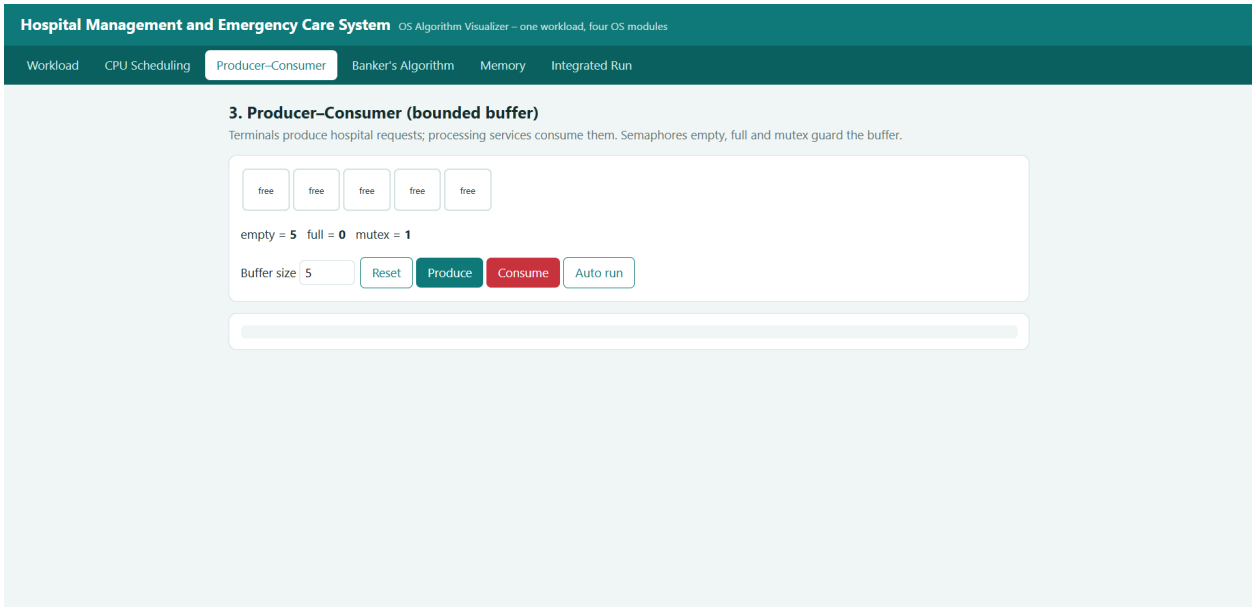
AIM: Mini Project -Design and Develop an Operating System Algorithm Visualizer





4. Deadlock avoidance – Banker's Algorithm

Resources: Imaging machine (A), Printer (B), DB connection (C). Available = total − allocation.

Allocation

	A	B	C
P0	1	1	0
P1	0	0	1
P2	1	0	1
P3	0	1	0
P4	1	0	1

Maximum

	A	B	C
P0	3	2	2
P1	1	1	1
P2	2	1	2
P3	1	2	1
P4	3	1	2

Need

	A	B	C
P0	2	1	2
P1	1	1	0
P2	1	1	1
P3	1	1	1
P4	2	1	1

Available: **[3,2,2]** Check safety

SAFE. Safe sequence: P0 → P1 → P2 → P3 → P4

```
P0: need [2,1,2] ≤ work [3,2,2] → finishes, releases [1,1,0]
P1: need [1,1,0] ≤ work [4,3,2] → finishes, releases [0,0,1]
P2: need [1,1,1] ≤ work [4,3,3] → finishes, releases [1,0,1]
P3: need [1,1,1] ≤ work [5,3,4] → finishes, releases [0,1,0]
P4: need [2,1,1] ≤ work [5,4,4] → finishes, releases [1,0,1]
```

Resource request

P0 Emergency Admission requests A 1 B 0 C 1 Request

Granted – state stays safe: P0 → P1 → P2 → P3 → P4

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5. Memory management

Each task is placed in a partition; leftover space inside a partition is shown as free.

Best Fit



Task	Size (KB)	Placed in
Emergency Admission	212	Partition 4 (300 KB)
Patient Registration	112	Partition 3 (200 KB)
Lab Report	417	Partition 2 (500 KB)
Pharmacy Request	90	Partition 1 (100 KB)
Imaging Task	426	Partition 5 (600 KB)

Total 1700 KB · used 1257 KB · free 443 KB · largest free block 174 KB

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